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Unit – 2 $\rightsquigarrow$ Probability

Introduction

- The world is governed by uncertainty.
 - Every second, countless events are unfolding, and no one knows exactly what will happen next.
- Yet, humans have always asked: "Can we predict the future? "
- We may never know the future with complete certainty, but mathematics gives us something almost as powerful:

The ability to measure uncertainty.

- That powerful tool is called **Probability**.
 - We cannot predict the future with certainty, but we can prepare for it with probability.*
- "Probability is the mathematics of uncertainty."
 - It helps us predict, analyze, and make smart decisions in an uncertain world.*

Why Probability is Important?

- Real Life Motivation
 - Probability helps us answer questions like:
 - Will it rain tomorrow?
 - What are the chances of getting placed?
 - Can AI predict diseases accurately?
 - What is the probability of network failure?
 - How risky is a stock market investment?
- Probability in Modern Technology
 - Probability forms the foundation of:
 - Artificial Intelligence & Machine Learning
 - Data Science
 - Cyber Security
 - Predictive Analytics

- Risk Analysis
- Probability powers modern intelligent systems.



- Probability theory is the branch of mathematics that is concerned with random (or chance) phenomena.
- It has attracted people to its study both because of its intrinsic interest and its successful applications to many areas within:
 - Physical Sciences
 - Biological Sciences
 - Social Sciences
 - Engineering
 - Business World
- The words **PROBABLE** and **POSSIBLE CHANCES** are quite familiar to us.
- We use these words when we are not sure of the result of certain events.
- There are two types of approaches in the theory of Probability:
 - Classical Approach
 - Introduced by Blaise Pascal
 - Axiomatic Approach
 - Introduced by A. Kolmogorov

Method – 1 $\rightsquigarrow$ Basic Terminologies

Introduction

- Probability theory begins with a few fundamental concepts which form the basic language of probability.
- Before studying probability laws and theorems, it is important to understand these terminologies clearly.

Random Experiment

- An experiment conducted under identical conditions is called a random experiment if it satisfies the following conditions:
 - It has more than one possible outcome.
 - All possible outcomes are known beforehand.
 - The exact outcome cannot be predicted with certainty in advance.
- Examples:
 - Tossing a coin **is** random experiment.
 - Throwing/rolling a die **is** random experiment.
 - Selecting a card from a pack of playing cards **is** random experiment.
 - Dividing 36 by 4 using a calculator **is not** random experiment.
- Identical conditions means
 - the method, equipment, and environment remain the same for every repetition of the experiment.
- Example: Coin Toss
 - Identical conditions mean:
 - Same coin
 - Same tossing method
 - Same landing surface
 - Only the outcome (Head or Tail) changes.
- The result of a random experiment is known as **outcome**.

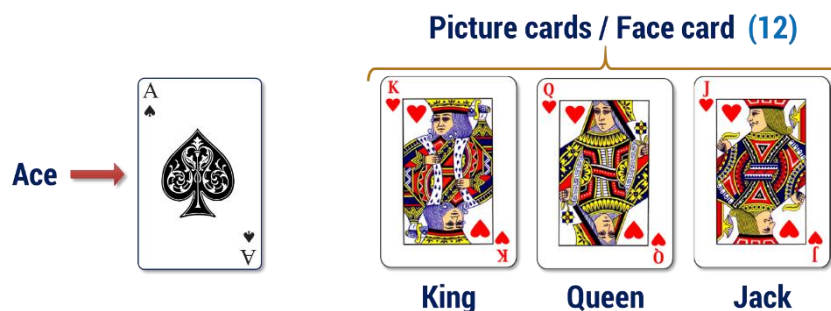
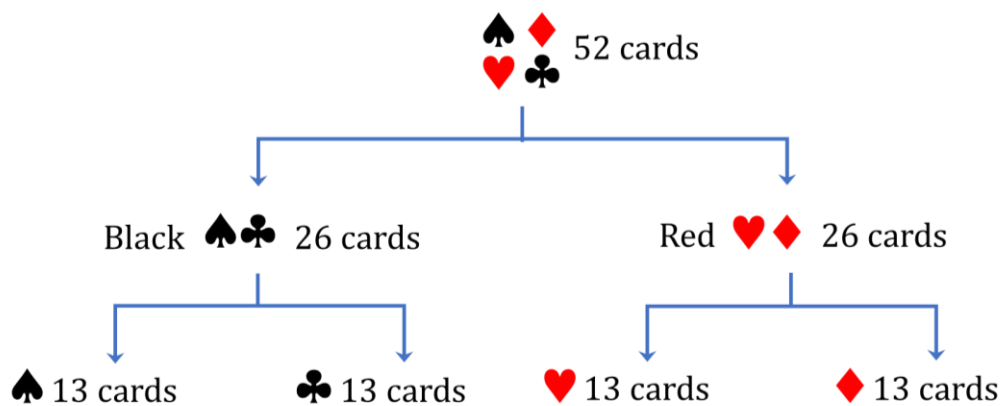
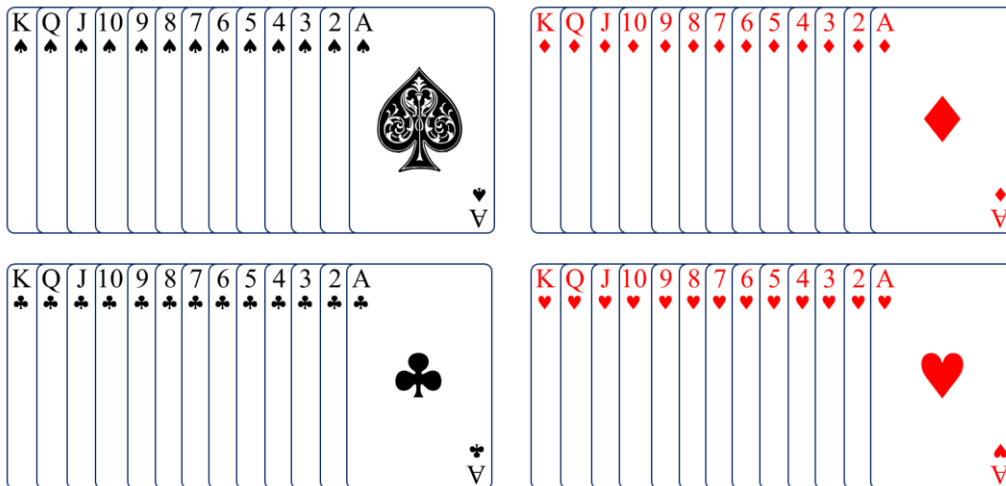
Unit – 2 Probability

Sample Space

- The **set of all possible outcome** is known as sample space of an experiment.
- Sample space is denoted by the symbol **S**.
- Elements of a sample space is known as **sample points**.
- **Playing Cards**

Deck of 52 playing cards which has four suits as follow:

SPADES ♠, CLUB ♣, DIAMOND ♦, HEART ♥



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Types of Sample Space

(1) Finite Sample Space

- A sample space containing **finite number of outcomes** is called finite sample space.
- Example: Tossing Two Coins

$$S = \{ HH, HT, TH, TT \}$$

- Number of outcomes = 4 (Finite)

(2) Infinite Sample Space

→ A sample space containing **infinite number of outcomes** is called infinite sample space.

→ Example 1: Toss a coin repeatedly until the first Head occurs.

$$S = \{ H, TH, TTH, TTTH, TTTTH, \dots \}$$

- Possible outcomes:
 - H (Head on the first toss)
 - TH (Tail then Head)
 - TTH (Two Tails then Head)
 - TTTH (Three Tails then Head)
 - ...
- The outcomes continue forever.
 - Hence, the sample space S is infinite.

→ Example 2: Suppose buses arrive randomly within the next 20 minutes.

- Sample Space: $S = \{ t \in \mathbb{R} : 0 \leq t \leq 20 \}$
- The waiting time could be
 - 5 min
 - 5.1 min
 - 5.12 min
 - 5.123456 min
 or any real number between 0 and 20.
- Hence, the sample space S is infinite.

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Types of Random Experiments

→ Random experiments are mainly classified into two types depending upon whether the selected item is returned back or not.

(1) Experiment **with** Replacement

→ An experiment is said to be with replacement if, after each selection, the selected object is returned to the original collection before the next selection.

→ In simple words,

- You select an item from a collection, **put it back**, and then select again.
- Each selection is independent, and the size and composition of the collection remain the same throughout the experiment.
- Example: 1
 - A bag contains 5 red balls and 3 blue balls.
 - One ball is selected and replaced before selecting the next ball.
 - Since the first ball is returned, the composition of the bag remains unchanged.
- Example: 2
 - A card is drawn from a deck of 52 cards, noted, and **returned** to the deck. The deck is then shuffled before drawing the next card.
 - Since the card is replaced, there are 52 cards available in each draw.
 - It is possible to draw the same card again.

(2) Experiment **without** Replacement

→ An experiment is said to be without replacement if the selected object is not returned to the original collection before the next selection.

→ In simple words:

- You select an item from a collection and **do not put it back** before selecting again.
- Each selection is dependent, and the size and composition of the collection change after every selection.

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→ Example: 1 A bag contains:

- 5 Red
- 3 Blue
 - One ball is drawn and **kept aside**.
 - If a red ball was drawn first, the bag now contains:
 - 4 Red
 - 3 Blue

→ Example: 2 A card is drawn from a deck of 52 cards and **not returned**.

- The second draw is made from **51 cards**.
- The first card cannot appear again.

→ **Summary**

With Replacement	Without Replacement
Select → Replace → Select Again	Select → Do Not Replace → Select Again
Same item may be selected again	Same item cannot be selected again
Collection (Sample Space) remains unchanged	Collection (Sample Space) changes after each selection
Selections are independent	Selections are dependent
Probabilities remain the same	Probabilities change after each selection

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→ A sample space contains all possible outcomes of a random experiment. However, in probability, we are usually interested in the occurrence of a particular outcome or a group of outcomes rather than the entire sample space. Such a subset of the sample space is called an event.

Event

→ Any **subset** of a **sample space** is known as an event.

→ Example:

- In experiment of Rolling a six-sided die and observing the number (or dots) that appear on top.
- A sample space of an experiment is
 $S = \{ 1, 2, 3, 4, 5, 6 \}$.
- Let A be an event that **even** number appears on top.
 $\therefore A = \{ 2, 4, 6 \}$
- Let B be an event that number **4** appears on top.
 $\therefore B = \{ 4 \}$
- Let C be an event that number **7** appears on top.
 $\therefore C = \phi$
- Let D be an event that number **less than 7** appears on top.
 $\therefore D = \{ 1, 2, 3, 4, 5, 6 \} = S$

Types of Events

→ **Impossible (Null) Event**

- An impossible event is an event that can never occur.
- The event E is known as impossible event if **$E = \phi$** .
- In previous example of rolling a die, event **$C = \phi$** therefore, it is impossible event.

→ **Certain (Sure) Event**

- An event that contains all the outcomes of the sample space is called a certain event.
i.e., The event E is known as sure event if **$E = S$**
- A certain event is an event that **always occurs**.

Unit – 2 Probability

- In previous example of rolling a die example, event $D = S$ therefore, it is sure event.

→ Simple or Elementary Event

- If an event E has **only one** sample point of a sample space, then it is known as a simple (or elementary) event.
- In previous example of rolling a die example, event B has only one sample point therefore, it is simple event.

→ Compound Event

- If an event E has **more than one** sample point of a sample space, then it is known as a compound event.
- In previous example of rolling a die example, event A and D are compound events.

→ Complementary Event

- The complement of an event A consists of all outcomes in the sample space that are not in A .
- Notation: A' (or $\bar{A}$)
- Note that $A' = S - A$.
- It is also known as the event “**not A**”.
- Example:
 - For experiment of tossing a coin twice, a sample space is
 $S = \{ HH, HT, TH, TT \}$
- Consider the following event:
 - Event A: **Exactly one head** appeared
 $\therefore A = \{ HT, TH \}$
 - The complementary event corresponds to event A is as follow:
 $\therefore A' = \{ HH, TT \}$

→ Mutually Exclusive Event [**Exclusive means not shared or kept separate**]

- Two or more events are said to be mutually exclusive if **no two events can occur simultaneously**.
- In other words, every pair of events has no common outcome.

Unit – 2 Probability

- Events $E_1, E_2, \dots, E_n$ of a sample space S are known as exhaustive events, if

$$E_i \cap E_j = \phi; i \neq j$$

→ The word **mutually** means with respect to each other.

- So, Mutually Exclusive means each event is exclusive with every other event.
- Example:
 - For experiment of throwing a die, a sample space is $S = \{1, 2, 3, 4, 5, 6\}$
 - $A = \{2, 4\}$, $B = \{1, 3\}$, $C = \{5, 6\}$ be events.
 - Now

$$A \cap B = \phi, \quad B \cap C = \phi, \quad C \cap A = \phi$$

- Since every pair of events has no common outcome, the events A, B and C are mutually exclusive

→ **Mutually Exhaustive Events** [Exhaustive means to cover everything completely]

- Two or more events are called mutually exhaustive if at least one of them always occurs. Together, they include all possible outcomes of the random experiment.
- Events $E_1, E_2, \dots, E_n$ of a sample space S are known as exhaustive events, if

$$\bigcup_{i=1}^n E_i = E_1 \cup E_2 \cup \dots \cup E_n = S$$

- Example:
 - In experiment of rolling a die, $S = \{1, 2, 3, 4, 5, 6\}$.
- Let us define the following events:
 - Event A : A number less than 4 appears
 $\therefore A = \{1, 2, 3\}$
 - Event B : A number greater than 2 but less than 5 appears
 $\therefore B = \{3, 4\}$
 - Event C : A number greater than 4 appears
 $\therefore C = \{5, 6\}$
 - Now,
 $A \cup B \cup C = \{1, 2, 3\} \cup \{3, 4\} \cup \{5, 6\}$

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$$= \{ 1, 2, 3, 4, 5, 6 \} = S$$

- Hence, events A, B and C are exhaustive events.

→ Mutually Exclusive and Exhaustive Events

- Two or more events are called mutually exclusive and exhaustive if they cannot happen together, and one of them must always happen.
- Furthermore, if

$$E_i \cap E_j = \phi \text{ for } i \neq j \text{ and}$$

$$\bigcup_{i=1}^n E_i = S,$$

then events $E_1, E_2, \dots, E_n$ are known as mutually exclusive and exhaustive events.

Example of Method-1: Basic Terminologies

C	1	A coin is tossed twice. The outcome of each toss is recorded. Write the sample space for the experiment. Answer: $S = \{ HH, HT, TH, TT \}$
H	2	Identify whether the following experiments are random or non-random. Give reason. (1) Tossing a coin (2) Rolling a die (3) Selecting a card from a deck (4) Switching ON a properly working electric bulb (5) Predicting tomorrow's weather
T	3	A basketball player attempts a free throw from a fixed distance. Discuss whether this experiment is random for: (1) A beginner player (2) A professional player

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H	4	State whether the following statements are true or false: (1) Every random experiment has more than one outcome. (2) Random experiment always gives same outcome. (3) Skill factor completely removes randomness. (4) Random experiment involves uncertainty.
H	5	Give two examples each of Random experiments and non-random experiments.
H	6	A football player attempts a penalty kick repeatedly under identical conditions. Is it a random experiment? Justify your answer.
T	7	Classify the following as random or non-random experiments: (1) Selecting a student randomly from a class (2) Calculating 5×7 (3) Measuring height of a student (4) Drawing a lottery ticket (5) Water boiling at 100°C
H	8	Find the sample space associated with the experiment of tossing a coin and rolling a die together.
T	9	Without writing the sample space explicitly: How many outcomes are possible when (1) A coin is tossed 5 times? (2) A die is rolled 4 times?
T	10	A balanced coin is tossed thrice. If three tails are obtained, a balance die is rolled. Otherwise, the experiment is terminated. Write down elements of the sample space. Answer: $S = \left\{ \begin{array}{l} \text{HHH, HHT, HTH, HTT, THH, THT, TTH,} \\ \text{TTT1, TTT2, TTT3, TTT4, TTT5, TTT6} \end{array} \right\}$

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C	11	Find the sample space associated with the experiment of rolling a pair of dice once. Also, find the number of elements of this sample space. Answer: $S = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6) \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6) \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6) \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6) \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6) \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) \end{array} \right\}$ Number of elements in sample space = $6 \times 6 = 6^2 = 36$
T	12	Three switches S_1, S_2, S_3 can each be either ON or OFF. Write the sample space of the experiment of observing the status of all three switches. Note: Denote: ON = 1 and OFF = 0
C	13	A coin is tossed. If it shows head, we draw a ball from a bag which contains 3 blue and 4 white balls; if it shows tail, we throw a die. Describe the sample space of this experiment. Answer: $S = \left\{ \begin{array}{l} HB_1, HB_2, HB_3, HW_1, HW_2, HW_3, HW_4, \\ T1, T2, T3, T4, T5, T6 \end{array} \right\}$
H	14	A die is rolled repeatedly until an even number appears on top for the first time. Describe the sample space.
T	15	A coin and die are tossed together until head and 4 appears. Write sample space.

C	16	The letters A, B, C and D are written separately on four slips of paper. The slips are put in a box and mixed thoroughly. A person draws two slips from the box, one after the other. Describe the sample space for the experiment in following cases: With replacement Without replacement Answer: (1) $S = \left\{ \begin{array}{l} AA, AB, AC, AD, BA, BB, BC, BD, \\ CA, CB, CC, CD, DA, DB, DC, DD, \end{array} \right\}$ (2) $S = \left\{ \begin{array}{l} AB, AC, AD, BA, BC, BD, \\ CA, CB, CD, DA, DB, DC \end{array} \right\}$
T	17	One die of red color, one of white color and one of blue color are placed in a bag. One die is selected at random and rolled, its color and the number on its uppermost face is noted. Describe the sample space. Answer: $S = \left\{ \begin{array}{l} R1, R2, R3, R4, R5, R6 \\ W1, W2, W3, W4, W5, W6 \\ B1, B2, B3, B4, B5, B6 \end{array} \right\}$
C	18	A box contains six balls numbered 1, 2, 3, 4, 5 and 6. Two balls are drawn successively without replacement, and the order of drawing is recorded. Write the sample space in set-builder form. Answer: $S = \{ (a, b) : a, b \in \{ 1, 2, 3, 4, 5, 6 \}, a \neq b \}$

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H	19	A coin is tossed 3 times. Give the elements of the following events: Event A: Getting at least two heads Event B: Getting exactly two tails Event C: Getting at most one tail Event D: Getting at least one tail Answer: $S = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$ Event A = $\{ HHH, HHT, HTH, THH \}$ Event B = $\{ HTT, THT, TTH \}$ Event C = $\{ HHH, HHT, HTH, THH \}$ Event D = $\{ HHT, HTH, HTT, THH, THT, TTH, TTT \}$
C	20	Consider the experiment of rolling an unbiased die. Write each event defined below and determine its type (impossible, sure, simple, or compound): Event E: The die shows a 4. Event F :The die shows an even number. Event G: The die shows a number greater than 7. Event H: The die shows a number less than or equal to 6. Answer: $S = \{ 1, 2, 3, 4, 5, 6 \}$ $E = \{ 4 \} \rightsquigarrow$ Simple event $F = \{ 2, 4, 6 \} \rightsquigarrow$ Compound event $G = \phi \rightsquigarrow$ Impossible event $H = \{ 1, 2, 3, 4, 5, 6 \} = S \rightsquigarrow$ sure event
C	21	In a random experiment, the total number of events is 128. (i) Find the number of elements in the sample space. (ii) How many simple events are possible? (iii) How many events are neither simple nor impossible? Answer: (i) $n = 7$, (ii) 7, (iii) 120

H	22	A positive integer is selected at random from the set of all divisors of 60. Let the events A: The selected number is even. B: The selected number is odd. C: The selected number is prime. D: The selected number is composite. E: The selected number is divisible by 3. F: The selected number is divisible by 5. G: The selected number is less than 10. H: The selected number is greater than 6. For each of the following pairs of events, determine whether they are Mutually Exclusive, Exhaustive, Both, or Neither. (i) A and B (ii) C and D (iii) E and F (iv) G and H Answer: (i) Both, (ii) Mutually Exclusive, (iii) Neither, (iv) Both
H	23	Two unbiased dice are thrown. Write down the following events: Event A: Both the dice show the same number. Event B: The sum of the numbers on the dice is 8. Event C: The sum of the number on the dice is any number from 2 to 6. Event D: The sum of the numbers on the dice is 13. Answer: A = { (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) } B = { (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) } C = { (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1) } D = ϕ

Unit – 2 Probability

H	24	Two unbiased dice are thrown. Write down the following events also determine type of each event: Event A: Both the dice show the same number. Event B: The total of the numbers on the dice is 12. Event C: The total of the numbers on the dice is 13. Event D: The total of the number on the dice is any number from 2 to 12. Answer: $A = \{ (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) \} \rightsquigarrow$ Compound event $B = \{ (6, 6) \} \rightsquigarrow$ Simple event $C = \phi \rightsquigarrow$ Impossible event $D = \{ (1, 1), \dots, (1, 6), \dots, (6, 1), \dots, (6, 6) \} = S \rightsquigarrow$ Sure event
H	25	A month of the year is selected at random. Write the elements of the following events: Event A: Month has 31 days. Event B: Month begins with letter J. Event C: Month has fewer than 31 days. Event D: Month has 32 days.
T	26	A fair die is tossed three times. Write the events corresponding to: A : Maximum outcome is 4. B : Minimum outcome is 3. C : All outcomes are less than 5. D : Exactly one outcome is greater than 4.

T	27	Committee X consists of 3 boys and 2 girls, while Committee Y consists of 2 boys and 4 girls. One member is selected at random from the two committees. Write the sample space and the elements of the following events: Event A: The selected member is a boy. Event B: The selected member is a girl. Event C: The selected member belongs to Committee X. Event D: The selected member belongs to Committee Y.
C	28	Consider the experiment of rolling a die. The following events are defined: A : getting a prime number B : getting an odd number. Write the sets representing the events (1) A or B (2) A and B (3) A but not B (4) not A Answer: (1) $A \cup B = \{ 1, 2, 3, 5 \}$, (2) $A \cap B = \{ 3, 5 \}$, (3) $A - B = \{ 2 \}$, (4) $A' = \{ 1, 4, 6 \}$
T	29	A password is formed using two characters chosen from the set {A, B, C}, where repetition is allowed. Let Event A: Password contains the letter A Event B: Password starts with B. Write the elements of following events: (3) A (4) B (5) $A' \cap B$ (6) $A \cap B$ (7) $A \cup B'$ (8) $B - A$

Unit – 2 Probability

Method – 2 $\rightsquigarrow$ Probability of an Event

Equally Likely Outcomes

- In a random experiment, equally likely outcomes are outcomes for which no outcome has a greater chance of occurring than the others.
 - In simple words, no outcome is favored over the others.
- Examples:
 - In a fair coin toss, Head and Tail are equally likely because neither outcome is favored.
 - In a fair die roll, each face (1 to 6) is equally likely because no face is favored.

Remarks

- Why "fair" and "unbiased"?
 - Fair means no outcome is favored over the others.
 - Biased means one or more outcomes have a greater chance of occurring than the others.
 - Unbiased simply means not biased, i.e., fair.
- Fair coin = Head and Tail are equally likely.
- Fair (unbiased) die = Each face from 1 to 6 is equally likely.
- **Note:** Unless otherwise specified, assume that a coin is fair and a die is fair (unbiased).
- Since the exact outcome of a random experiment cannot be predicted in advance, we need a way to measure the chance of an event occurring.
- This measure is called **Probability**.

Classical Definition of Probability

- Probability is a numerical measure of the chance that an event will occur in a random experiment.

OR

- Probability is a number that represents how likely an event is to occur in a random experiment.

Unit – 2 Probability

→ When all outcomes of a random experiment has a **finite sample space** and all outcomes are **equally likely**, the probability of an event E, denoted by P(E), is defined as:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

or

$$P(E) = \frac{\text{number of elements in event E}}{\text{number of elements in sample space S}} = \frac{n(E)}{n(S)} ; E \subseteq S$$

→ Example:

- One card is drawn from a well-shuffled deck of 52 cards.

The probability of getting a **queen** = $\frac{1}{52}$

- A box contains 3 blue, 2 white, and 4 red marbles. A marble is drawn at random from the box, what is the probability that it will be white?

Solution:

$$S = \{ B_1, B_2, B_3, W_1, W_2, R_1, R_2, R_3, R_4 \}$$

$$\text{Number of possible outcomes} = 3 + 2 + 4 = 9$$

$$\text{i.e., } n(S) = 9$$

Let W be the event that marble is **white**.

$$W = \{ W_1, W_2 \}$$

$$\text{i.e., } n(W) = 2$$

$$\text{Therefore, } P(W) = \frac{n(W)}{n(S)} = \frac{2}{9}$$

Basic Properties of Probability

(1) For any event A, $0 \leq P(A) \leq 1$.

- Probability measures the chance of an event occurring.
- Probability can be expressed as a fraction, decimal, or percentage.
- Therefore,
 - $P(A) < 0$ is not possible, since a probability cannot be negative.
 - $P(A) > 1$ (or greater than 100%) is not possible, since no event can be more certain than a certain event.

Unit – 2 Probability

In classical probability,

(2) $P(A) = 0 \Leftrightarrow A$ is an impossible event

Proof:

- An impossible event has no favourable outcomes.
- Therefore, $n(A) = 0$ and

$$P(A) = \frac{n(A)}{n(S)} = \frac{0}{n(S)} = 0$$

(3) $P(A) = 1 \Leftrightarrow A$ is a sure event.

→ **Proof:**

- A sure event contains all possible outcomes.
- Therefore, $n(A) = n(S)$ and

$$P(A) = \frac{n(A)}{n(S)} = \frac{n(S)}{n(S)} = 1$$

Some Important Results

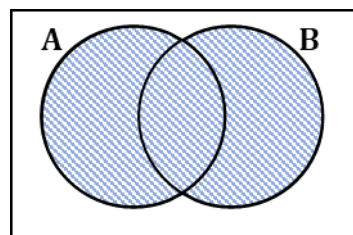
(1) $P(A') = 1 - P(A)$ **or** $P(A) = 1 - P(A')$

(2) If A and B are mutually exclusive events, $P(A \cap B) = 0$.

(3) If A and B are mutually exhaustive events, $P(A \cup B) = 1$

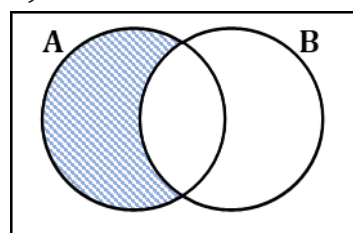
(4) The probability that **at least one** out of the events A and B will occur is,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



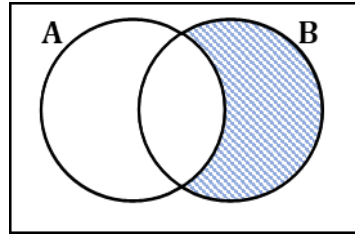
(5) The probability that **only event A** out of the events A and B will occur is,

$$P(A \cap B') = P(A) - P(A \cap B)$$



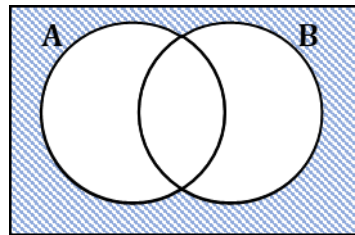
- (6) The probability that **only event B** out of the events A and B will occur is,

$$P(A' \cap B) = P(B) - P(A \cap B)$$



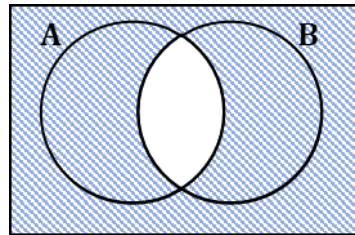
- (7) The probability that **none** of the event A and B occur is,

$$P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B) \text{ (De Morgan's Rule)}$$



- (8) The probability that events A and B **not occur together** is,

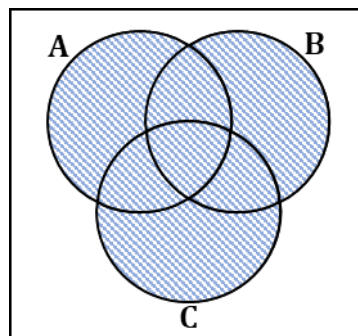
$$P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B) \text{ (De Morgan's Rule)}$$



- (9) The probability that **at least one** of the events A, B and C will occur is,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C)$$

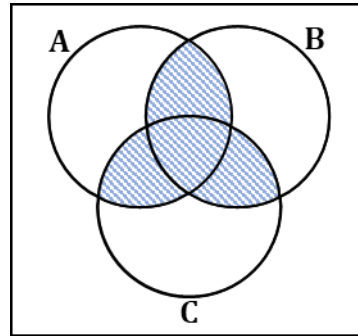
$$+ P(A \cap B \cap C)$$



Unit – 2 Probability

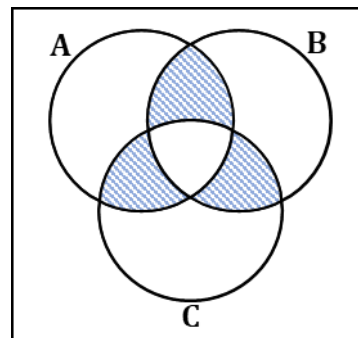
(10) The probability that **at least two** of the three events occur is,

$$P[(A \cap B) \cup (B \cap C) \cup (C \cap A)] = P(A \cap B) + P(B \cap C) + P(A \cap C) - 2P(A \cap B \cap C)$$



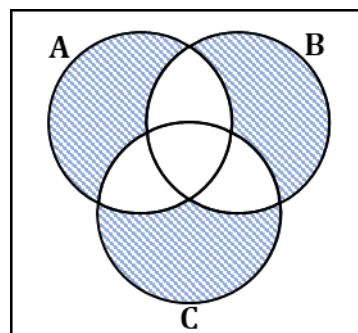
(11) The probability that **exactly two** of the three events occur is,

$$P[(A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A' \cap B \cap C)] \\ = P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$$



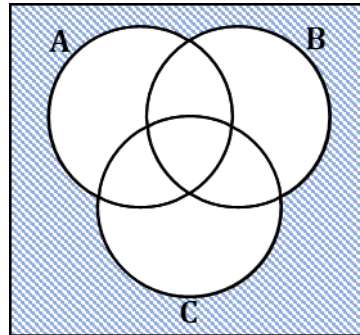
(12) The probability that **exactly one** of the three events occur is,

$$P[(A \cap B' \cap C') \cup (A' \cap B \cap C') \cup (A' \cap B' \cap C)] \\ = P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(B \cap C) - 2P(A \cap C) + 3P(A \cap B \cap C)$$



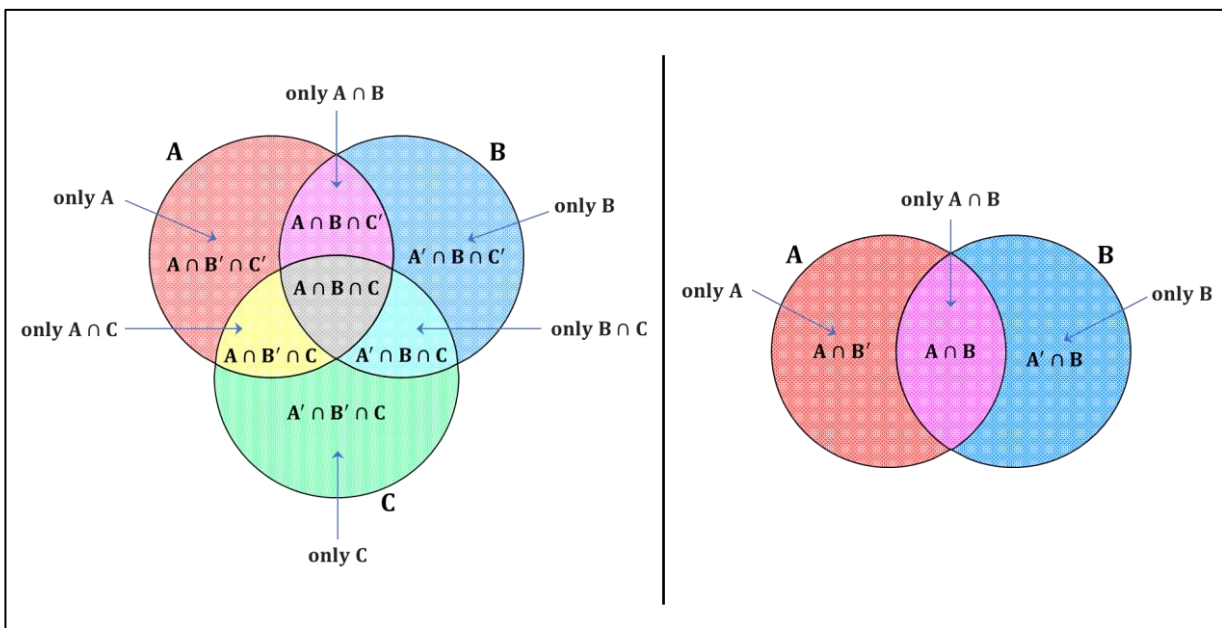
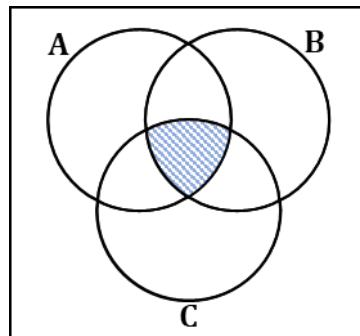
(13) The probability that **none** of the event A, B and C occur is,

$$P(A' \cap B' \cap C') = P(A \cup B \cup C)' = 1 - P(A \cup B \cup C) \text{ (De Morgan's Rule)}$$



(14) The probability that events A, B and C **not occur together** is,

$$P(A' \cup B' \cup C') = P(A \cap B \cap C)' = 1 - P(A \cap B \cap C) \text{ (De Morgan's Rule)}$$



Example of Method-2: Probability of an Event

C	1	Two unbiased dice are thrown. Find the probability that: (1) The total of the numbers on the dice is greater than 8. (2) The total of the numbers on the dice is 13. (3) Total of numbers on the dice is any number from 2 to 12, both inclusive. Answer: (1) $\frac{5}{18}$ (2) 0 (3) 1
T	2	In a group of 1000 persons, there are 650 who can speak Hindi, 400 can speak English, and 150 can speak both Hindi and English. If a person selected at random, what is the probability that a person speaks (1) Hindi only, (2) English only, (3) Only one of two languages, (4) At least one of the two languages. Answer: (1) 0.5 (2) 0.25 (3) 0.75 (4) 0.9
T	3	A person applies for a job in two firms A and B, the probability of his being selected in firm A is 0.7 and being rejected in firm B is 0.5. The probability of at least one of the applications being rejected is 0.6. What is the probability that person will be selected in one of the two firms? Answer: 0.8
H	4	Three newspapers A, B, C are published in a certain city. It is estimated from a survey that of the adult population: 20% read A, 16% read B, 14% read C, 8% read both A and B, 5% read both A and C, 4% read B and C, 2% read all three. Find what percentage read at least one of the papers? Answer: 35%
C	5	An urn contains 6 green, 4 red and 9 black balls. If 3 balls are drawn at random, find the probability that at least one is green. Answer: $\frac{683}{969}$

Unit – 2 Probability

C	6	If A and B are two mutually exclusive events with $P(A) = 0.30$, $P(B) = 0.45$. Find the probability of A', $A \cap B$, $A \cup B$, $A' \cap B'$. Answer: $P(A') = 0.7$ $P(A \cap B) = 0$ $P(A \cup B) = 0.75$ $P(A' \cap B') = 0.25$
H	7	Two unbiased dice are thrown. Find the probability that: (1) Both the dice show the same number. (2) The first die shows 6. (3) The total of the numbers on the dice is 8. (4) The total of the numbers on the dice is divisible by 2 or 3. Answer: (1) $\frac{1}{6}$ (2) $\frac{1}{6}$ (3) $\frac{5}{36}$ (4) $\frac{2}{3}$
C	8	If 5 cards are drawn from a pack of 52 well-shuffled cards, find the probability of (1) 4 ace (2) 4 aces and 1 is a king (3) 3 are tens and 2 are jacks (4) a nine, ten, jack, queen, king is obtained in any order (5) 3 are of any one suit and 2 are of another (6) at least one ace is obtained Answer: (1) $\frac{1}{54145}$ (2) $\frac{1}{649740}$ (3) $\frac{1}{108290}$ (4) $\frac{64}{162435}$ (5) $\frac{429}{4165}$ (6) $\frac{18472}{54145}$
T	9	Four cards are drawn from the pack of cards. Find the probability that (1) All are diamonds (2) There is one card of each suit (3) There are two spades and two hearts (4) All are red or all are picture (face) cards. Answer: (1) $\frac{11}{4165}$ (2) $\frac{2197}{20825}$ (3) $\frac{468}{20825}$ (4) $\frac{3089}{54145}$

Unit – 2 Probability

T	10	4 cards are drawn at random from a pack of 52 cards. Find probability that (1) They are a king, a queen, a jack and an ace. (2) Two are kings and two are queens. (3) Two are black and two are red. (4) Two cards of hearts and two cards of diamonds. Answer: (1) $\frac{256}{270725}$ (2) $\frac{36}{270725}$ (3) $\frac{325}{833}$ (4) $\frac{468}{20825}$
T	11	A box contains 5 red, 6 white and 2 black balls. The balls are identical in all aspects other than color. (1) One ball is drawn at random from the box. Find the probability that the selected ball is black. (2) Two balls are drawn at random from the box. Find the probability that one ball is white and one is red. Answer: (1) $\frac{2}{13}$ (2) $\frac{5}{13}$
T	12	If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls. What is the probability that one of the balls is white and the other two black? Answer: 0.3636
T	13	There are 5 yellow, 2 red, and 3 white balls in the box. Three balls are randomly selected from the box. Find the probability of the following events. (1) All balls are of different color (2) 2 yellow and 1 red color ball (3) all balls are of same color. Answer: (1) 0.25 (2) 0.1667 (3) 0.0917
T	14	A box contains 6 red balls, 4 white balls, 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color. Answer: 0.5275

Unit – 2 Probability

T	15	A room has three lamp sockets. From a collection of 10 light bulbs of which only 6 are good. A person selects 3 at random and puts them in the socket. What is the probability that the room will have light? Answer: $\frac{29}{30}$
C	16	Consider a poker hand of five cards. Find the probability of getting four of a kind (i.e., four cards of the same face value) assuming the five cards are chosen at random. Answer: $\frac{1}{4165}$
C	17	An integer is chosen at random from the first 200 positive integers. What is the probability that the integer is divisible by 6 or 8? Answer: 0.25
C	18	A basket contains 20 apples and 10 oranges of which 5 apples and 3 oranges are bad. If a person takes 2 at random, what is the probability that either both are apples or both are good? Answer: $\frac{316}{435}$
C	19	Do as directed: (1) Find the probability that there will be 5 Sundays in the month of July. (2) Find the probability that there will be 5 Sundays in the month of June. (3) What is the probability that a non-leap year contains 53 Sundays? (4) What is the probability that a leap year contains 53 Sundays? Answer: (1) $\frac{3}{7}$ (2) $\frac{2}{7}$ (3) $\frac{1}{7}$ (4) $\frac{2}{7}$
C	20	Four letters of the word THURSDAY are arranged in all possible ways. Find the probability that the word formed is HURT. Answer: $\frac{1}{1680}$

Unit – 2 Probability

T	21	A four-digit number is formed using the digits 0, 1, 2, 3, 4 without repetition. Find the probability that the number formed is even. Answer: $\frac{5}{8}$
T	22	A single die is tossed once. Find the probability of a 2 or 5 turning up. Answer: $\frac{1}{3}$
H	23	Three coins are tossed. Find the probability of (1) Getting at least 2 heads. (2) Getting exactly 2 heads. Answer: (1) 0.5 (2) 0.375
T	24	A card is drawn from a pack of 52 cards. Find the probability of getting a king or a heart or a red card. Answer: $\frac{7}{13}$
H	25	One card is drawn at random from a well shuffled pack of 52 cards. Find probability that the card will be an ace, a card of black color, a diamond, and not an ace. Answer: 0.0769, 0.5, 0.25, 0.9231
T	26	A machine produces a total of 12000 bolts a day, which are on the average 3% defective. Find the probability that out 600 bolts chosen at random, 12 will be defective. Answer: $\frac{\binom{360}{12} \binom{11640}{588}}{\binom{12000}{600}}$
T	27	If 5 of 20 tyres in storage are defective and 5 of them are randomly chosen for inspection (that is, each tire has the same chance of being selected), what is the probability that the two of the defective tires will be included? Answer: 0.2935

Unit – 2 Probability

T	28	A class has 10 boys and 5 girls. Three students are selected at random one after the other. Find the probability that (1) First two are boys and third is girl. (2) First and third of same gender and second is of opposite gender. Answer: (1) $\frac{15}{91}$ (2) $\frac{5}{21}$
T	29	A survey was conducted among 200 people to study their interest in two sports, Cricket and Football. It was found that 120 people like Cricket, 90 people like Football, and 50 people like both Cricket and Football. If one person is selected at random from the survey, find the probability that the selected person: (1) Only Cricket. (2) Only Football. (3) At least one of the two games. (4) Neither Cricket nor Football. (5) Cricket and Football do not occur together. Answer: (1) 0.35 (2) 0.20 (3) 0.80 (4) 0.20 (5) 0.75
T	30	A survey was conducted among 400 students regarding their participation in three sports: Cricket, Football, and Volleyball. The survey revealed that 180 students play Cricket, 150 play Football, 120 play Volleyball, 70 play both Cricket and Football, 60 play both Football and Volleyball, 50 play both Cricket and Volleyball, and 30 play all three games. If one student is selected at random, find the probability that the selected student: (1) Plays at least one of the three games. (2) Plays at least two of the three games. (3) Plays exactly two of the three games. (4) Plays exactly one of the three games. (5) Plays none of the three games. (6) Does not play all three games. Answer: (1) 0.75 (2) 0.30 (3) 0.225 (4) 0.45 (5) 0.25 (6) 0.925

Method – 3 $\Rightarrow$ Conditional Probability

Introduction

- In many practical situations, the occurrence of one event affects the occurrence of another event.
- The probability of an event may change when some additional information about the experiment becomes available.
- Such probabilities are called conditional probabilities.
- Consider the following situations:
 - Probability that a student gets placed **given that** the student knows programming.
 - Probability of rain **given that** the sky is cloudy.
 - Probability of disease **given that** a medical test result is positive.
 - Probability of network failure **given that** server traffic is unusually high.
 - Probability that an email is spam **given that** it contains suspicious keywords.
- In all these situations, some prior information is already known. Therefore, the probability changes according to the given condition.
- Example:
 - Imagine a college seminar in an auditorium filled with **180** students.
 - 60 of them study AI/ML.
 - Out of those 60 AI/ML students, 35 live in the hostel.
 - Question 1:
 - “What is the sample space if we select one student randomly?”

Since we know absolutely nothing about the student yet, we have to look at the whole room. Our total pool is all 180 students.
 - Question 2:
 - “What is the chance the student is a hosteller, given that they study AI/ML?”

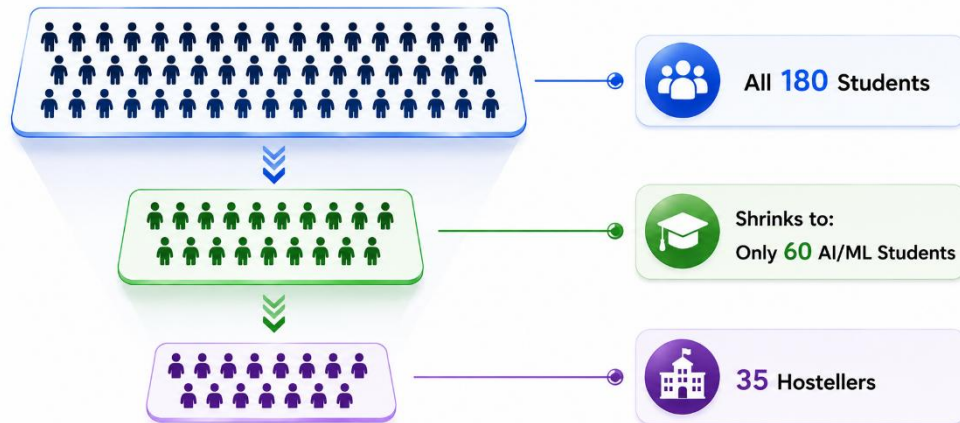
The phrase “given that” means the clue is already a fact.

We know the student is in the AI/ML class.

Unit – 2 Probability

Therefore, the remaining students outside the AI/ML group are no longer considered in the sample space.

Our focus instantly shrinks:



Dependent Events

→ Two events are said to be dependent if the occurrence or non-occurrence of one event changes the probability of occurrence of the other event.

→ Example:

- Suppose **two cards** are drawn **one after another** from a pack of 52 cards **without replacement**.
 - If the first card drawn is an ace, then only 51 cards remain in the deck.
 - Thus, probability of drawing an ace in the second draw changes.
 - Hence, the events are dependent.

Conditional Probability

→ Let A and B be any two events of a finite sample space S. The probability of occurrence of an event A under the condition that event B has already occurred is called the conditional probability.

→ It is denoted as **$P(A | B)$** and read as “probability of A given B” and defined as follow:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

→ Similarly, “probability of B given A” is defined as follow:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) \neq 0$$

Unit – 2 Probability

→ Example:

- Consider the experiment of tossing three fair coins.
 - To find probability of “getting at least two heads given that first coin shows tail” is conditional probability.

The sample space of the experiment is

$$S = \{ HHH, HHT, HTH, THH, HTT, THT, TTH, TTT \}.$$

Let E and F denote the following events :

E : at least two head appear

$$E = \{ HHH, HHT, HTH, THH \}$$

F : first coin shows tail

$$F = \{ THH, THT, TTH, TTT \}$$

Here, event F has occurred before event E.

$$\text{Now, to find } P(E | F) = \frac{P(E \cap F)}{P(F)}$$

$$E \cap F = \{ THH \}$$

$$\text{Thus, } P(F) = \frac{4}{8} = \frac{1}{2} \text{ and } P(E \cap F) = \frac{1}{8}$$

$$\text{Therefore, } P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{(1/8)}{(1/2)} = \frac{1}{4}$$

→ Properties of Conditional Probability

- Let A_1, A_2 and B be any three events of a sample space S, then

$$P(A_1 \cup A_2 | B) = P(A_1 | B) + P(A_2 | B) - P(A_1 \cap A_2 | B); P(B) > 0.$$
- Let A and B be any two events of a sample space S, then

$$P(A' | B) = 1 - P(A | B); P(B) > 0.$$

Unit – 2 Probability

Multiplicative Law of Probability

- Let A and B be any two events in a finite sample space S. The rule used to determine the probability of **simultaneous occurrence** of events A and B is called the Multiplicative Law of Probability.
- The probability of simultaneous occurrence of events A and B is given by:
 - $P(A \cap B) = P(A) \cdot P(B | A); P(A) \neq 0$
 $= P(B) \cdot P(A | B); P(B) \neq 0$
- For A, B and C be any three events in the sample space S,
 - $P(A \cap B \cap C) = P(A) \cdot P(B | A) \cdot P(C | A \cap B)$

Independent Events

- Two events are said to be independent if the occurrence or non-occurrence of one event does not change the probability of occurrence of the other event.
- Let A and B be two events in a finite sample space S.
- If the probability of occurrence of event A does not change due to occurrence (or non-occurrence) of event B, then event A and B are called independent events.

i.e., $P(A | B) = P(A)$ and $P(B | A) = P(B)$

- Example:
 - In an auditorium, let
 A = Event that a randomly selected student belongs to AI/ML specialization
 B = Event that the auditorium projector is functioning properly
 - The occurrence of event A does not affect the occurrence of event B and vice versa. Therefore, A and B are independent events.

→ Important Results

- Two events A and B are independent if and only if
 $P(A \cap B) = P(A) \cdot P(B)$
Note: This result is **commonly used to verify** whether two events are independent.
- If A and B are independent events, then
 - The events A' and B' are also independent. Hence,
 $P(A' \cap B') = P(A') \cdot P(B')$.

Unit – 2 Probability

- The events A and B' are also independent. Hence, $P(A \cap B') = P(A') \cdot P(B')$.
- The events A' and B are also independent. Hence, $P(A' \cap B) = P(A') \cdot P(B)$.

Mutually Independent Events

→ The events A, B, and C are said to be mutually independent if the occurrence or non-occurrence of any event does not affect the probability of occurrence of the other events.

→ The events A, B, and C are said to be **mutually independent** if

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(B \cap C) = P(B) \cdot P(C)$$

$$P(A \cap C) = P(A) \cdot P(C)$$

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

→ If the last condition is not satisfied, the events are said to be pairwise independent.

Note: If A, B, and C are mutually independent, then their complementary events are also mutually independent.

Example of Method-3: Conditional Probability and Independent Events

C	1	$P(A) = \frac{1}{3}$, $P(B') = \frac{1}{4}$, $P(A \cap B) = \frac{1}{6}$, then find $P(A \cup B)$, $P(A' \cap B')$ and $P(A' B')$. Answer: $P(A \cup B) = \frac{11}{12}$ $P(A' \cap B') = \frac{1}{12}$ $P(A' B') = \frac{1}{3}$
C	2	If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(A \cup B) = \frac{1}{2}$, then find $P(B A)$, $P(A B')$. Answer: $P(B A) = \frac{1}{4}$ $P(A B') = \frac{1}{3}$
C	3	If A and B are independent events, with $P(A) = \frac{3}{8}$, $P(B) = \frac{7}{8}$. Find $P(A \cup B)$, $P(A B)$ and $P(B A)$. Answer: $P(A \cup B) = \frac{59}{64}$ $P(A B) = \frac{3}{8}$ $P(B A) = \frac{7}{8}$

C	4	A bag contains 5 red balls and 3 blue balls. Two balls are drawn one after another without replacement. Find the probability that: (1) The second ball is red, given that the first ball drawn is red. (2) The second ball is blue, given that the first ball drawn is red. Answer: (1) $\frac{4}{7}$ (2) $\frac{3}{7}$																				
C	5	A person is known to hit the target in 3 out of 4 shots, whereas another person is known to hit the target in 2 out of 3 shots. What is probability that target will be hit? Answer: $\frac{11}{12}$																				
C	6	A market survey was conducted in four cities to find out the preference for brand X soap. The responses are shown below: <table><tr><td></td><td>Delhi</td><td>Kolkata</td><td>Chennai</td><td>Mumbai</td></tr><tr><td>Yes</td><td>45</td><td>55</td><td>60</td><td>50</td></tr><tr><td>No</td><td>35</td><td>45</td><td>35</td><td>45</td></tr><tr><td>No opinion</td><td>5</td><td>5</td><td>5</td><td>5</td></tr></table> (1) What is the probability that a consumer preferred brand X, given that he was from Chennai? (2) Given that a consumer preferred brand X, what is the probability that he was from Mumbai? Answer: (1) 0.6, (2) 0.23		Delhi	Kolkata	Chennai	Mumbai	Yes	45	55	60	50	No	35	45	35	45	No opinion	5	5	5	5
	Delhi	Kolkata	Chennai	Mumbai																		
Yes	45	55	60	50																		
No	35	45	35	45																		
No opinion	5	5	5	5																		

Unit – 2 Probability

H	7	A problem in statistics is given to three students A, B and C, whose chances of solving it independently are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that (1) the problem is solved (2) at least two of them are able to solve the problem (3) exactly two of them are able to solve the problem (4) exactly one of them is able to solve the problem Answer: (1) $\frac{3}{4}$, (2) $\frac{7}{24}$, (3) $\frac{1}{4}$, (4) $\frac{11}{24}$
C	8	In a group of 200 students 40 are taking English, 50 are taking Mathematics, 12 are taking both. (1) If a student is selected at random, what is the probability that the student is taking English? (2) A student is selected at random from those taking Mathematics. What is the probability that the student is taking English? (3) A student is selected at random from those taking English, what is the probability that the student is taking Mathematics? Answer: (1) 0.20, (2) 0.24, (3) 0.3
C	9	A bag contains 6 white, 9 black balls. 4 balls are drawn at a time. Find the probability for first draw to give 4 white & second draw to give 4 black balls in each of following cases: (1) The balls are replaced before the second draw. (2) The balls are not replaced before the second draw. Answer: (1) $\frac{6}{5915}$, (2) $\frac{3}{715}$
H	10	If $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$ and $P(A \cup B) = \frac{11}{12}$. Find $P(A B)$. Answer: $P(A B) = \frac{2}{9}$

Unit – 2 Probability

T	11	For two independent events A & B if $P(A) = 0.3$ and $P(A \cup B) = 0.6$. Find $P(B)$. Answer: 0.4286
H	12	If A, B are independent events and $P(A) = \frac{1}{4}$, $P(B) = \frac{2}{3}$. Find $P(A \cup B)$. Answer: 0.75
T	13	Check weather events A and B are independent or not if $P(A) = 0.20$, $P(B) = 0.40$ and $P(A \cup B) = 0.50$. Answer: Not Independent
H	14	If A and B are independent events with $P(A) = 0.26$, $P(B) = 0.45$, find $P(A \cap B)$, $P(A \cap B')$, $P(A' \cap B')$. Answer: 0.117, 0.143, 0.407
T	15	In producing screws, let A mean “screw is too slim” and B “screw is too short”. Let $P(A) = 0.1$ and $P(B \cap A) = 0.02$. A screw, selected randomly, is of type A, what is probability that a screw is of type B? Answer: 0.2
T	16	A problem in statistics is given to three students A, B, C whose chances of solving it are 0.5, 0.75 and 0.25 respectively. What is the probability that the problem will be solved if all of them try independently? Answer: 0.90625
T	17	In a box, 100 bulbs are supplied out of which 10 bulbs have defects of type A, 5 bulbs have defects of type B and 2 bulbs have defects of both the type. Find the probability that, (1) A bulb to be drawn at random has a B – type defect under the condition that it has an A – type defect. (2) A bulb to be drawn at random has no B – type defect under the condition that it has no A – type defect. Answer: (1) 0.2, (2) 0.9667

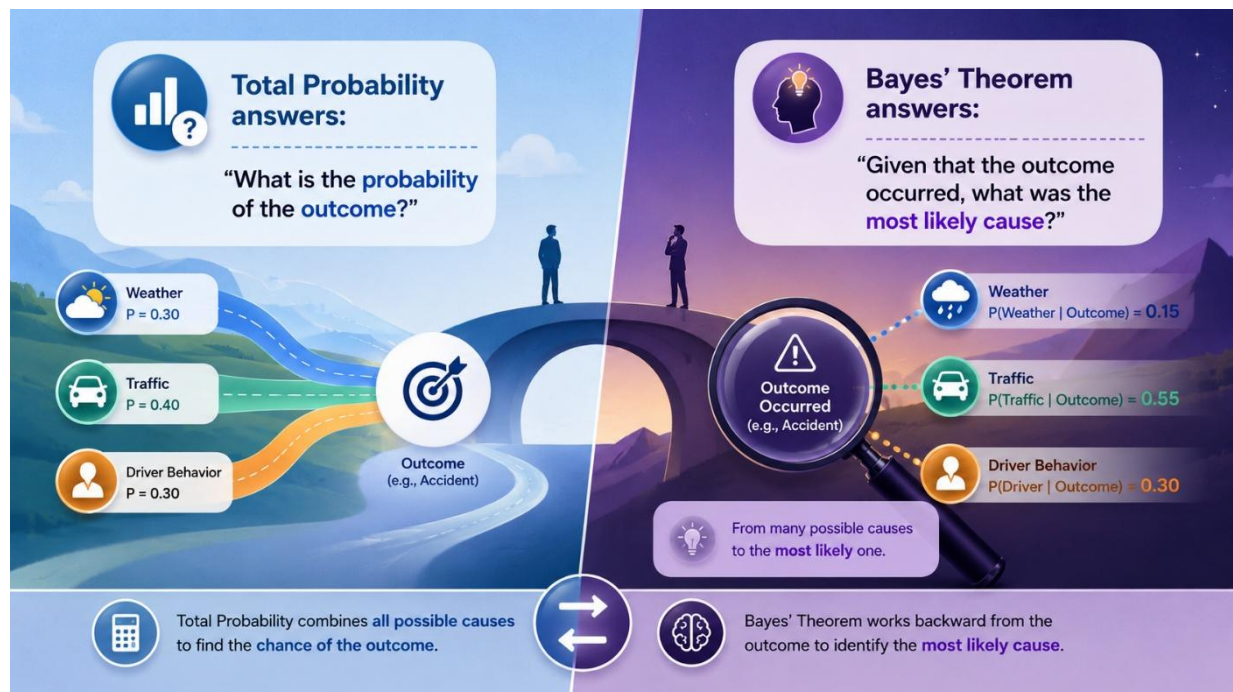
Unit – 2 Probability

T	18	In a certain college 25% of the students failed in probability and 15% of the student failed in statistics and 10% of the students failed in both. A student is selected at random, if he failed in probability, what is probability that he failed in statistics? Answer: 0.4
T	19	Two integers are selected at random from 1 to 11. If the sum is even, find the probability that both the integers are odd. Answer: 0.6
T	20	A card is drawn from a well-shuffled deck of 52 cards and then second card is drawn, find the probability that one card is a spade and then second card is club if the first card is not replaced. Answer: $\frac{13}{204}$
T	21	From a bag containing 4 white and 6 black balls, two balls are drawn at random. If the balls are drawn one after the other without replacements, find the probability that one is white and one is black. Answer: $\frac{8}{15}$
?	22	A QR code scanner successfully scans a QR code with probability 0.85 on a single attempt. If two independent scanning attempts are made, find the probability that the QR code is scanned successfully at least once. What does your answer reveal about the reliability of making two scanning attempts? Answer: 97.75%

T	23	An autonomous drone verifies the delivery location using two independent sensors: GPS verification succeeds with probability 0.94. Camera-based verification succeeds with probability 0.90. The package is delivered only if both verifications are successful. Compute (1) Probability that the package is delivered successfully. (2) Probability that at least one verification fails. (3) Given that GPS verification is successful, find the probability that camera verification is also successful. (4) Interpret your answer. What does it reveal about the reliability of the delivery verification system? Answer: (1) 0.846 (2) 0.154 (3) 0.90
T	24	A smartphone uses a face recognition system for unlocking. During testing, it is observed that the system correctly recognizes the owner's face in 96% of the attempts. If the owner tries to unlock the phone, find the probability that the phone fails to recognize the owner, given that the attempt is made by the owner. Answer: 4%

Method – 4 $\rightarrow$ Bayes' Theorem

Introduction



- In many real-world situations, an outcome can be influenced by several possible factors.
- Before the outcome occurs, we are interested in finding the overall chance that it will happen.
 - Consider a road accident. The accident may be caused by:
 - Weather conditions
 - Traffic conditions
 - Driver behavior
 - Each factor contributes differently to the occurrence of the accident. To determine the overall probability of an accident, we must consider all possible contributing factors and combine their effects.
- Thus, **Total Probability** answers the question:

"What is the probability of the outcome?"
- In many real-world situations, an outcome has already occurred and we are interested in identifying the most likely factor responsible for it.

Unit – 2 Probability

→ After the outcome occurs, our focus shifts from finding the probability of the outcome to identifying the most likely cause behind it.

- Consider a road accident. The accident may have been caused by:
 - Weather conditions
 - Traffic conditions
 - Driver behaviour
- Since several factors can contribute to the occurrence of an accident, the observed accident can be used to determine which factor was most likely responsible for it.

→ Thus, **Bayes' Theorem** answers the question:

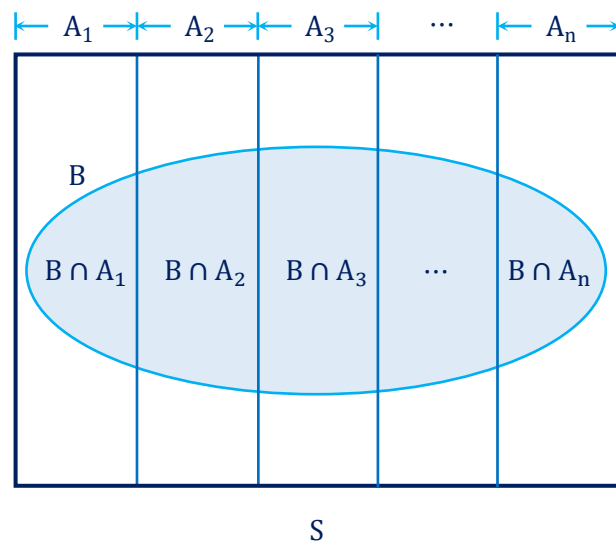
"Given that the outcome occurred, what was the most likely cause?"

Total Probability

→ Let $A_1, A_2, \dots, A_n$ be **mutually exclusive and exhaustive** events of the sample space S with $P(A_i) \neq 0$, for $i = 1, 2, \dots, n$. Let B be any event associated with S . Then, the probability of B is

$$P(B) = P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2) + \dots + P(A_n) \cdot P(B | A_n)$$

- Explanation:



- Using multiplicative law of probability, we get
- $P(B \cap A_i) = P(A_i) \cdot P(B | A_i), \forall i = 1, 2, \dots, n$ (' $\forall$ ' means **for every**)

Unit – 2 Probability

- Thus,
- $P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_i)$
- $P(B) = P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2) + \dots + P(A_n) \cdot P(B | A_n)$

→ Example:

- Factory A and Factory B, that produce electronic components. Factory A produces 60% of the components and among those, 10% are defective. Factory B produces the remaining 40% of the components and among those, 5% are defective.
- Find the probability that a randomly selected electronic component is defective.

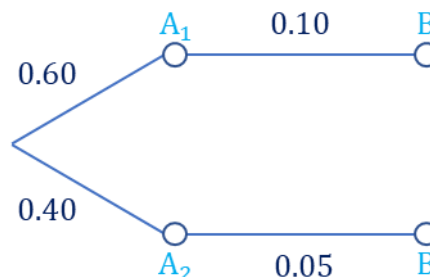
Solution:

- Let A_1 : The component is from Factory A.

A_2 : The component is from Factory B.

B : The component is defective.

Since every component is produced either by Factory A or Factory B, the events A_1 and A_2 are mutually exclusive and exhaustive.



Therefore,

$$P(A_1) = 60\% = 0.60 \quad P(B | A_1) = 10\% = 0.10$$

$$P(A_2) = 40\% = 0.40 \quad P(B | A_2) = 5\% = 0.05$$

Thus,

$$\begin{aligned}
 P(B) &= P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2) \\
 &= (0.60) \cdot (0.10) + (0.40) \cdot (0.05) \\
 &= 0.06 + 0.02 \\
 &= 0.08
 \end{aligned}$$

Unit – 2 Probability

Bayes' Theorem

→ Let $A_1, A_2, A_3, \dots, A_n$ be **mutually exclusive and exhaustive** events of the sample space S with $P(A_i) \neq 0$, for $i = 1, 2, 3, \dots, n$. Let B be any event associated with S with $P(B) \neq 0$. The probability of an event A_i when the event B has occurred is

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2) + \dots + P(A_n) \cdot P(B | A_n)}$$

Example of Method-4: Total Probability and Bayes' Theorem

C	1	In a certain assembly plant, three machines, B_1, B_2 and B_3 , make 30%, 45% and 25%, respectively, of the products. It is known from the past experience that 2%, 3% and 2% of the products made by each machine respectively are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective? Answer: 0.0245
T	2	State Bayes' theorem. In a bolt factory, three machines A, B and C manufacture 25%, 35% and 40% of the total product respectively. Out of these outputs 5%, 4% and 2% respectively, are defective bolts. A bolt is picked up at random and found to be defective. What are the probabilities that it was manufactured by machine A, B and C? Answer: 0.3623, 0.4058, 0.2319
H	3	A microchip company has two machines that produce the chips. Machine-I produces 65% of the chips, but 5% of its chips are defective. Machine-II produces 35% of the chips, but 15% of its chips are defective. A chip is selected at random and found to be defective. What is the probability that it came from Machine-I? Answer: 0.3824

Unit – 2 Probability

T	4	In a computer engineering class, 5% of the boys and 10% of the girls have an IQ of more than 150. In this class, 60% of student are boys. If a student is selected random and found to have IQ more than 150, find the probability that the student is a boy. Answer: $\frac{3}{7}$
C	5	An urn contains 10 white and 3 black balls, while another urn contains 3 white and 5 black balls. Two balls are drawn from the first urn and put into the second urn and then a ball is drawn from the later. What is the probability that it is a white ball? Answer: $\frac{59}{130}$
C	6	Consider two boxes, first with 5 green and 2 pink and second with 4 green and 3 pink balls. Two balls are selected from random box. If both balls are pink, find the probability that they are from second box. Answer: $\frac{3}{4}$
T	7	Urn A contain 1 white, 2 black, 3 red balls; Urn B contain 2 white, 1 black, 1 red balls; Urn C contain 4 white, 5 black, 3 red balls. One urn is chosen at random & two balls are drawn. These happen to be one white & one red. What is probability that they come from urn A? Answer: 0.2797
C	8	A company has two plants to manufacture hydraulic machine. Plant I manufacture 70% of the hydraulic machines and plant II manufactures 30%. At plant I, 80% of hydraulic machines are rated standard quality and at plant II, 90% of hydraulic machine are rated standard quality. A machine is picked up at random and is found to be of standard quality. What is the chance that it has come from plant I? Answer: 0.6747

Unit – 2 Probability

C	9	If proposed medical screening on a population, the probability that the test correctly identifies someone with illness as positive is 0.99 and the probability that test correctly identifies someone without illness as negative is 0.95. The incident of illness in general population is 0.0001. You take the test the result is positive then what is the probability that you have illness? Answer: 0.002
H	10	There are three boxes, Box - I contains 10 light bulbs of which 4 are defective, Box - II contains 6 light bulbs of which 1 is defective and Box - III contains 8 light bulbs of which 3 are defective. A box is chosen and a bulb is drawn. Find the probability that the bulb is defective. Answer: 0.3139
H	11	Suppose that the population of a certain city is 40% male & 60% female. Suppose also that 50% of male & 30% of female smokes. Find the probability that a smoker is male. Answer: $\frac{10}{19}$
C	12	A card from a pack of 52 cards is lost. From the remaining cards of pack, two cards are drawn and are found to be hearts. Find the probability of the missing card to be a heart. Answer: $\frac{11}{50}$
T	13	Suppose there are three chests each having two drawers. The first chest has a gold coin in each drawer, the second chest has a gold coin in one drawer and a silver coin in the other drawer and the third chest has a silver coin in each drawer. A chest is chosen at random and a drawer opened. If the drawer contains a gold coin, what is the probability that the other drawer also contains a gold coin? Answer: $\frac{2}{3}$

Unit – 2 Probability

T	14	An insurance company insured 2000 bike drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a bike driver, a car driver and a truck driver is 0.10, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a bike driver? Answer: 0.1639
H	15	Three hospitals contain 10%, 20% and 30% of diabetes patients. A Patient is selected at random who is diabetes patient. Determine the probability that this patient comes from first hospital. Answer: 0.1667
T	16	Three persons A, B and C have applied for a job in a private company. The chance of their selections is in the ratio 1 : 2 : 4. The probabilities that A, B and C can introduce changes to improve the profits of the company are 0.8, 0.5 and 0.3, respectively. If the change does not take place, find the probability that it is due to the appointment of C. Answer: 0.7
T	17	A computer network is monitored using an Intrusion Detection System (IDS). The probability that an intrusion occurs is 0.2%. The IDS correctly detects an intrusion 98% of the time. However, it also gives a false alarm 1% of the time when there is no intrusion. If the IDS reports an intrusion, what is the probability that an intrusion has actually occurred? Answer: 16.42%
T	18	A smartphone fingerprint scanner correctly recognizes the owner 97% of the time and incorrectly accepts another person's fingerprint 2% of the time. Only 1 in 500 unlock attempts is made by an unauthorized user. If the phone unlocks successfully, find the probability that it was unlocked by an unauthorized user. Answer: 0.00413%

T	19	An autonomous vehicle uses an AI vision system to detect obstacles on the road. Obstacles are actually present in 1% of the captured frames. The vision system correctly detects an obstacle with probability 0.98. It falsely reports an obstacle in 2% of the frames where no obstacle exists. If the system reports an obstacle, find the probability that an obstacle is actually present. Interpret your answer. What does it reveal about the reliability of the obstacle detection system? Answer: 33.11%
T	20	A satellite monitors forest regions for wildfires. On any day, only 0.4% of the monitored regions actually contain a wildfire. The satellite correctly detects a wildfire with probability 0.97. It incorrectly reports a wildfire in 1.5% of regions where no wildfire exists. A region is reported to have a wildfire. (1) Find the probability that the region actually has a wildfire. (2) Interpret your answer. What does it reveal about the reliability of satellite-based wildfire detection? Answer: 20.62%

Method – 5 $\Rightarrow$ Random Variables

Introduction

- In many random experiments, the outcomes are not always numbers.
- For example, when a coin is tossed, the outcomes are Head (H) and Tail (T).
- To represent such outcomes numerically, we assign a number to each outcome.
- This numerical assignment is called a **Random Variable**.

Random Variable

- Let S be a sample space of a random experiment. Every element of S need not always be a number. However, we wish to assign a specific number to each outcome.
- A Random Variable is a **function** that **associates a real number** with **each outcome** of the sample space of a random experiment.
- Random variables are often denoted by X, Y.
- That is, if S is the sample space, then a random variable is a function

$$X : S \rightarrow \mathbb{R}$$

- where $\mathbb{R}$ denotes the set of **real numbers**.
- Random variables are classified into **two** types:
 - (1) **Discrete** random variable
 - (2) **Continuous** random variable

Discrete Random Variable

- A random variable that can take only **finite values or countably infinite** values is known as a **Discrete Random Variable**.
- The values of a discrete random variable can be **listed** or **counted**.
- Example:

If two coins are tossed simultaneously the following sample space is generated:

$$S = \{ HH, HT, TH, TT \}$$

Let X denotes the number of heads obtained.

$$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0$$

Thus, random variable X can take values 0, 1, 2.
- Some other examples of discrete random variable.

Unit – 2 Probability

- Number of children in a family.
- Numbers of stars in the sky.
- Number of defective items in a batch.

Continuous random variable:

- A random variable that can take all values within a given interval is known as a Continuous Random Variable.
- The values of a continuous random variable are obtained by measurement and **cannot be listed** individually.
- Example:
 - The age of a person
 - Weight and height of a person
 - Life of an electric bulb

Probability Distribution of random variable

- The probability distribution of a random variable describes how probabilities are distributed among its possible values.
- If X is a **random variable** and x is one of its **possible values**, then the **probability** assigned to x is denoted by $P(X = x)$.

Probability Function

- A probability distribution can often be represented mathematically by a function.
- If X is a random variable and $f(x)$ is a real-valued function such that $P(X = x) = f(x)$, then $f(x)$ is called the Probability Function of the random variable X .
- The function $f(x)$ gives the probability corresponding to the value x of the random variable X .
- Probability functions can be classified as
 - (1) Probability Mass Function (P. M. F.) (for discrete random variable)
 - (2) Probability Density Function (P. D. F.) (for continuous random variable)

Unit – 2 Probability

Probability Mass Function

→ The probability function $f(x_i) = P(X = x_i) = p(x_i)$ of a **discrete** random variable X is called the **probability mass function**, if it satisfies following conditions:

$$(1) \quad 0 \leq p(x_i) \leq 1, \text{ for all } i$$

$$(2) \quad \sum_{i=1}^n p(x_i) = 1$$

Note:

Let X be a discrete random variable taking values

$$a < x_1 < x_2 < \cdots < x_k < \cdots < x_{n-1} < x_n < b,$$

then

$$P(X < b) = P(X = a) + P(X = x_1) + \cdots + P(X = x_{n-1}) + P(X = x_n)$$

$$P(x_1 \leq X \leq x_k) = P(X = x_1) + P(X = x_2) + \cdots + P(X = x_k)$$

$$P(X > a) = P(X = x_1) + P(X = x_2) + \cdots + P(X = b)$$

Probability Density Function

→ The probability function $f(x)$ of a **continuous** random variable X is called the **probability density function**, if it satisfies following conditions:

$$(1) \quad f(x) \geq 0, \text{ for all } x \in \mathbb{R}$$

$$(2) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

Note:

- Let a and b be any two real numbers such that $a < b$. Then,

$$P(x \leq b) = \int_{-\infty}^b f(x) dx$$

$$P(a < x < b) = \int_a^b f(x) dx$$

$$P(a < x) = \int_a^{\infty} f(x) dx$$

Unit – 2 Probability

Remark:

- For a **continuous random variable**, the probability at any particular value is always zero,
i.e., $P(X = c) = 0$, for all $c \in \mathbb{R}$
- Hence, $P(a < x < b) = P(a \leq x < b) = P(a < x \leq b) = P(a \leq x \leq b)$

Example of Method-5: Random Variable and probability Function

C	1	Is $P(X = x) = \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{1-x}$; $x = 0, 1$ a probability function? Answer: Yes																		
H	2	Is $P(X = x) = \left(-\frac{1}{2}\right)^x$; $x = 0, 1, 2$ a probability function? Answer: No																		
C	3	A random variable X has the following probability function. Find the value of k and then evaluate $P(x < 6)$, $P(x \geq 6)$ and $P(0 < x < 5)$. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$P(X = x)$</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr></table> Answer: $k = 0.1$ $P(x < 6) = 0.81$ $P(x \geq 6) = 0.19$ $P(0 < x < 5) = 0.8$	X	0	1	2	3	4	5	6	7	$P(X = x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$
X	0	1	2	3	4	5	6	7												
$P(X = x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$												
T	4	From a lot of 8 bulbs, including 3 defective bulbs, two bulbs are drawn at random. Let X denote the number of defective bulbs selected. Construct the probability distribution of X. Answer: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$P(X = x)$</td><td>$\frac{10}{28}$</td><td>$\frac{15}{28}$</td><td>$\frac{3}{28}$</td></tr></table>	X	0	1	2	$P(X = x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$										
X	0	1	2																	
$P(X = x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$																	

Unit – 2 Probability

T	5	Find k for the probability distribution $p(x) = k \binom{4}{x}$, $x = 0, 1, 2, 3, 4$. Answer: $k = \frac{1}{16}$										
C	6	If $P(X = x) = \frac{x}{15}$, $x = 1$ to 5. Find $P(1 \text{ or } 2)$ & $P(0.5 < X < 2.5 \mid X > 1)$. Answer: $P(1 \text{ or } 2) = \frac{1}{5}$ $P(0.5 < X < 2.5 \mid X > 1) = \frac{1}{7}$										
T	7	Probability distribution of a random variable X is given below. Find $P(2 \leq x \leq 4)$, $P(x > 2)$, $P(x \text{ is odd})$ and $P(x \text{ is even})$ <table border="1"><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$P(X = x)$</td><td>0.1</td><td>0.2</td><td>0.5</td><td>0.2</td></tr></table> Answer: $P(2 \leq x \leq 4) = 0.9$ $P(x > 2) = 0.7$ $P(x \text{ is odd}) = 0.6$ $P(x \text{ is even}) = 0.4$	X	1	2	3	4	$P(X = x)$	0.1	0.2	0.5	0.2
X	1	2	3	4								
$P(X = x)$	0.1	0.2	0.5	0.2								
C	8	Verify that the following function is P.D.F or not? $f(x) = \begin{cases} \frac{2x}{9} \left(2 - \frac{x}{2}\right) & ; 0 \leq x \leq 3 \\ 0 & ; \text{elsewhere} \end{cases}$ Answer: Yes										
T	9	Find the constant c such that the function $f(x) = \begin{cases} cx^2 & ; 0 < x < 3 \\ 0 & ; \text{elsewhere} \end{cases}$ is a probability density function and Compute $P(1 < X < 2)$, $P(X \leq 2)$, $P(X \geq 2)$. Answer: $c = \frac{1}{9}$ $P(1 < X < 2) = \frac{7}{27}$ $P(X \leq 2) = \frac{8}{27}$ $P(X \geq 2) = \frac{19}{27}$										

Unit – 2 Probability

C	10	Is the function $f(x)$ defined as below is a probability density function? If so, find the probability that the variate having this density falls in the interval $(1, 2)$. $f(x) = \begin{cases} e^{-x} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases}$ Answer: Yes, $P(1 < X < 2) = 0.2325$
T	11	Check whether the following function $f(x) = \begin{cases} \frac{3 + 2x}{18} & ; 2 \leq x \leq 4 \\ 0 & ; \text{otherwise} \end{cases}$ is a probability density function? If yes, then find $P(3 \leq X \leq 4)$. Answer: Yes $P(3 \leq X \leq 4) = \frac{5}{9}$
H	12	If $P(x) = \frac{2x + 1}{48}$, $x = 1, 2, 3, 4, 5, 6$. Verify whether $P(x)$ is probability function or not. Answer: Yes
T	13	Find k for the probability distribution $p(x) = k \binom{4}{x}$, $x = 0, 1, 2, 3, 4$. Answer: $k = \frac{1}{16}$
T	14	Verify that the following function is P.D.F or not? $f(x) = \begin{cases} \frac{x}{8} & ; 0 \leq x < 2 \\ \frac{1}{4} & ; 2 \leq x < 4 \\ \frac{6 - x}{8} & ; 4 \leq x < 6 \end{cases}$ Answer: Yes

Unit – 2 Probability

- T 15** A smart traffic signal records the number of vehicles waiting at a junction when the signal turns green. The observations for 200 signal cycles are given below.

Number of vehicles (<u>X</u>)	0	1	2	3	4	5
Frequency	12	28	50	56	38	16

- (1) Construct the probability distribution of X.
 (2) Verify that it is a valid probability distribution.
 (3) Find:
 (a) $P(X \geq 3)$
 (b) $P(1 \leq X < 4)$
 (c) $P(X \text{ is even})$

Answer:

(1)

X	0	1	2	3	4	5
P(X = x)	0.06	0.14	0.25	0.28	0.19	0.08

(2) Yes

(3) $P(X \geq 3) = 0.55$ $P(1 \leq X < 4) = 0.67$ $P(X \text{ is even}) = 0.50$

Method – 6 $\rightsquigarrow$ Various Measures of Statistics

Introduction

- A probability distribution describes the possible values of a random variable and their probabilities.
- To summarize and analyze a probability distribution, certain numerical measures are used.
- These measures are called Various Measures of Statistics.
- The most important measures are:
 - (1) Mathematical Expectation (Mean)
 - (2) Variance
 - (3) Standard Deviation

Mathematical Expectation

- The mathematical expectation of a random variable is the **weighted average** of its possible values, where the **weights** are their corresponding **probabilities**.

→ For a Discrete Random Variable

- Let X be a **discrete** random variable having various values $x_1, x_2, \dots, x_n$ with probability mass function $p(x_i) = P(X = x_i)$. The mathematical expectation of X, denoted by **E(X)**, is defined by

$$E(X) = \sum_{i=1}^n x_i \cdot p(x_i).$$

→ For a Continuous Random Variable

- Let X be a **continuous** random variable with probability density function **f(x)**. The mathematical expectation of X, denoted by **E(X)**, is defined by

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx.$$

- **E(X)** is also known as the **expected value** or **mean** of the random variable X.

Unit – 2 Probability

Properties of Mathematical Expectation

- (1) For any constant **c**, $E(c) = c$.
 (2) If **a**, **b** and **c** are constants ($c \neq 0$), then

$$E\left(\frac{aX \pm b}{c}\right) = \frac{1}{c} [a \cdot E(X) \pm b]$$

- (3) If X and Y are two random variables, then $E(X + Y) = E(X) + E(Y)$.
 (4) If X and Y are two **independent** random variables, then $E(X \cdot Y) = E(X) \cdot E(Y)$.

Expectation of X^2

- The expectation of the square of a random variable X is denoted by $E(X^2)$.
 → **For a Discrete Random Variable**

$$E(X^2) = \sum_{i=1}^n x_i^2 \cdot p(x_i)$$

- **For a Continuous Random Variable**

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$$

Note: The quantity $E(X^2)$ is frequently used in the calculation of variance of a random variable.

Variance of Random Variable:

- Variance is a characteristic of a random variable and is used to measure the dispersion (or variation) of its values about the mean.
 → Let X be a random variable with mean $E(X)$. The expected value of the squared deviation of X from its mean ($E(X)$) is called the variance of X and is denoted by **V(X)**.

$$\text{i.e., } V(X) = E[(X - E(X))^2]$$

$$\text{Thus, } V(X) = E(X^2) - [E(X)]^2$$

Properties of Variance

- (1) For any constant **c**, $V(c) = 0$.
 (2) If **a** and **b** are constants, then

$$V(aX + b) = a^2 \cdot V(X)$$

 (3) If X and Y are the **independent** random variables, then $V(X + Y) = V(X) + V(Y)$.

Unit – 2 Probability

Standard Deviation of Random Variable

→ The positive **square root of $V(X)$** (Variance of X) is called **standard deviation** of random variable X and is denoted by σ .

$$\sigma = \sqrt{V(X)}$$

Example of Method-6: Various Measures of Statistics

C	1	Probability distribution of a random variable X is given below. Find $E(X)$, $V(X)$, $\sigma(X)$, $E(3X + 2)$, $V(3X + 2)$. <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$P(X = x)$</td><td>0.1</td><td>0.2</td><td>0.5</td><td>0.2</td></tr></table> Answer: $E(X) = 2.8$, $V(X) = 0.76$, $\sigma(X) = 0.8718$, $E(3X + 2) = 10.4$, $V(3X + 2) = 6.84$	X	1	2	3	4	$P(X = x)$	0.1	0.2	0.5	0.2						
X	1	2	3	4														
$P(X = x)$	0.1	0.2	0.5	0.2														
C	2	Let mean and standard deviation of a random variable X be 5 & 5 respectively, find $E(X^2)$ and $E(2X + 5)^2$. Answer: $E(X^2) = 50$, $E(2X + 5)^2 = 325$																
C	3	The following table gives the probabilities that a certain computer will malfunction 0, 1, 2, 3, 4, 5 or 6 times on any one day. <table><tr><td>Number of malfunctions x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Probability $p(x)$</td><td>0.17</td><td>0.29</td><td>0.27</td><td>0.16</td><td>0.07</td><td>0.03</td><td>0.01</td></tr></table> Find the mean and variance of this probability distribution. Answer: Mean = 1.8, Variance = 1.8	Number of malfunctions x	0	1	2	3	4	5	6	Probability $p(x)$	0.17	0.29	0.27	0.16	0.07	0.03	0.01
Number of malfunctions x	0	1	2	3	4	5	6											
Probability $p(x)$	0.17	0.29	0.27	0.16	0.07	0.03	0.01											
C	4	4 raw mangoes are mixed accidentally with the 16 ripe mangoes. Find the probability distribution of the raw mangoes in a draw of 2 mangoes. Answer: $P(X = 0) = \frac{12}{19}$, $P(X = 1) = \frac{32}{95}$, $P(X = 2) = \frac{3}{95}$																

Unit – 2 Probability

C	5	There are 3 red and 2 white balls in a box and 2 balls are taken at random from it. A person gets Rs. 20 for each red ball and Rs. 10 for each white ball. Find the expected amount won by the player. Answer: 32												
C	6	A random variable X has P. D. F $f(x) = \begin{cases} \frac{3 + 2x}{18} & ; 2 \leq x \leq 4 \\ 0 & ; \text{otherwise} \end{cases}$. Find the standard deviation of the distribution. Answer: 0.5726												
T	7	The probability distribution of a random variable X is given below. Find a, E(X), E(2X + 3), E(X² + 2), V(X), V(3X + 2). <table><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>P(X)</td><td>$\frac{1}{12}$</td><td>$\frac{1}{3}$</td><td>a</td><td>$\frac{1}{4}$</td><td>$\frac{1}{6}$</td></tr></table> Answer: $a = \frac{1}{6}$, $E(X) = \frac{1}{12}$, $E(2X + 3) = \frac{19}{6}$, $E(X^2 + 2) = \frac{43}{12}$, $V(X) = \frac{227}{144}$, $V(3X + 2) = \frac{227}{16}$	X	-2	-1	0	1	2	P(X)	$\frac{1}{12}$	$\frac{1}{3}$	a	$\frac{1}{4}$	$\frac{1}{6}$
X	-2	-1	0	1	2									
P(X)	$\frac{1}{12}$	$\frac{1}{3}$	a	$\frac{1}{4}$	$\frac{1}{6}$									
H	8	The probability distribution of a random variable X is given below. <table><tr><td>X</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>$\frac{3}{10}$</td><td>$\frac{1}{10}$</td><td>k</td><td>$\frac{3}{10}$</td><td>$\frac{1}{10}$</td></tr></table> Find k, E(X), E (4X + 3), E(X²), V(X), V(2X + 3). Answer: $k = \frac{1}{5}$, $E(X) = \frac{4}{5}$, $E (4X + 3) = \frac{31}{5}$, $E(X^2) = \frac{13}{5}$, $V(X) = \frac{49}{25}$, $V(2X + 3) = \frac{196}{25}$	X	-1	0	1	2	3	P(X)	$\frac{3}{10}$	$\frac{1}{10}$	k	$\frac{3}{10}$	$\frac{1}{10}$
X	-1	0	1	2	3									
P(X)	$\frac{3}{10}$	$\frac{1}{10}$	k	$\frac{3}{10}$	$\frac{1}{10}$									
T	9	Three balanced coins are tossed, find the mathematical expectation of tails. Answer: $\frac{3}{2}$												

Unit – 2 Probability

H	10	(1) A contestant tosses a coin and receives \$5 if head appears and \$1 if tail appears. What is the expected value of a trial? (2) A contestant receives \$4.00 if a coin turns up heads and pays \$3.00 if it turns tails. What is the expected value of a trail? Answer: \$ 3.00, \$ 0.50
T	11	A machine produces on average of 500 items during first week of the month & average of 400 items during the last week of the month. The probability for these being 0.68 and 0.32. Determine the expected daily production. Answer: 468
T	12	In a business, the probability that a trader can get profit of Rs. 5000 is 0.4 and probability for loss of Rs. 2000 is 0.6. Find his expected gain or loss. Answer: 800
H	13	There are 8 apples in a box, of which 2 are rotten. A person selects 3 Apples at random from it. Find the expected value of the rotten apples. Answer: 0.75
H	14	There are 10 bulbs in a box, out of which 4 are defectives. If 3 bulbs are taken at random, find the expected number of defective bulbs. Answer: 1.2
T	15	A random variable X has pdf $f(x) = kx^2(4 - x)$; $0 < x < 4$. Find the value of k and hence find its mean and standard deviation. Answer: $k = \frac{3}{64}$, Mean = 2.4, SD = 0.8
T	16	A random variable X has pdf $f(x) = kx^2(1 - x^3)$; $0 < x < 1$. Find the value of k and hence find its mean and variance. Answer: $k = 6$, Mean = $\frac{9}{14}$, SD = $\frac{9}{245}$

T**17**

The duration (in minutes) for which the internet connection is unavailable is represented by the following probability distribution.

X (minutes)	2	4	6	8	10
P(X)	0.12	0.18	0.30	0.25	0.15

Let X denote the duration of the internet service interruption on a randomly selected day. The compensation (in rupees) paid to customers is given by $C = 120X^2 + 30X + 50$.

Calculate

- (1) The expected duration of the internet service interruption.
- (2) The variance of the duration.
- (3) The standard deviation of the duration.
- (4) The expected compensation paid to the customers.

Answer: (1) 6.26 minutes (2) 5.9724 min²

(3) 2.4438 minutes (4) ₹ 5657

T**18**

A company is evaluating two Internet Service Providers, ISP-A and ISP-B, for providing internet connectivity to its corporate office. The number of successful internet sessions established during a randomly selected hour is represented by the random variables X and Y, respectively. The probability distributions are given below.

X	40	50	60	70	80
P(X = x)	0.10	0.20	0.40	0.20	0.10
Y	30	40	60	80	90
P(Y = y)	0.15	0.20	0.30	0.20	0.15

- (1) Find the mean, variance, and standard deviation of the number of successful internet sessions for each ISP.
- (2) Compare the average performance of the two ISPs.
- (3) Which ISP provides a more consistent service?
- (4) If you were the network administrator, which ISP would you recommend for the corporate office? Justify your answer using the statistical measures obtained.

Answer: (1) $E(X) = 60$ $V(X) = 120$ $\sigma(X) = 10.9545$

$E(Y) = 60$ $V(Y) = 430$ $\sigma(Y) = 20.7364$

(2) Both ISPs provide the same average performance.

(3) ISP – A provides a more consistent service.

(4) ISP – A is recommended for the corporate office.

Method – 7 $\rightsquigarrow$ Cumulative Distribution Function

Introduction

- The probability function and probability density function describe the probability distribution of a random variable.
- In many applications, it is useful to determine the probability that a random variable assumes a **value less than or equal to** a specified number.
- This leads to the concept of the **Cumulative Distribution Function (CDF)**, which gives the cumulative probability up to a given value.

Cumulative Distribution Function

- The **cumulative distribution function (CDF)** of a random variable X is denoted by **$F(x)$** and is defined as

$$F(x) = P(X \leq x), \quad -\infty < x < \infty$$

- A cumulative distribution function is **also called** the **distribution function**.
- Sometimes the notation **$F_X(x)$** is used when it is **necessary to specify the random variable**.
- **For a Discrete Random Variable**
 - Let X be a **discrete** random variable taking the values $x_1, x_2, \dots$ with probability mass function **$p(x_i) = P(X = x_i)$** .
 - Assume that $x_1 < x_2 < \dots < x_k < \dots$
 - The distribution function **$F(x)$** of the discrete random variable X is defined by

$$F(x_k) = P(X \leq x_k) = \sum_{i=1}^k p(x_i)$$

The set of pairs $\{ (x_i, F(x_i)) \}$, $i = 1, 2, \dots$ is known as the **cumulative probability distribution**.

x_i	x_1	x_2	...
$F(x_i)$	$p(x_1)$	$p(x_1) + p(x_2)$	...

Unit – 2 Probability

→ For a **Continuous** Random Variable

- Let X be a **continuous** random variable with probability density function $f(x)$.
- The distribution function $F(x)$ of the continuous random variable X is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt ; -\infty < x < \infty$$

Properties of Cumulative Distribution Function

→ Let $F(x)$ be the cumulative distribution function (CDF) of a random variable X .

- For all $x \in \mathbb{R}$, $0 \leq F(x) \leq 1$
- $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$
- If $x_1 < x_2$, $F(x_1) \leq F(x_2)$
i.e., $F(x)$ is a **non-decreasing** function.
- $P(X > x) = 1 - F(x)$
- $P(X < x) = F(x) - P(X = x)$
- $P(X \geq x) = 1 - F(x) + P(X = x)$
- $P(a < X \leq b) = F(b) - F(a)$
- $P(a \leq X \leq b) = F(b) - F(a) + P(X = a)$
- $P(a < X < b) = F(b) - F(a) - P(X = b)$
- $P(a \leq X < b) = F(b) - F(a) + P(X = a) - P(X = b)$

- Remark (Continuous Random Variable)**

If X is a continuous random variable, then

$$P(X = a) = 0 \text{ and } P(X = b) = 0,$$

Hence,

$$P(a < x < b) = P(a \leq x < b) = P(a < x \leq b) = P(a \leq x \leq b) = F(b) - F(a)$$

(11) **Additional Property (Continuous Random Variable)**

- If X is a continuous random variable with probability density function $f(x)$, then

$$F'(x) = f(x); \quad f(x) \geq 0$$

Example of Method-6: Cumulative Distribution Function

C	1	A random variable X takes the values $-3, -2, -1, 0, 1, 2, 3$ such that $P(X = 0) = P(X > 0) = P(X < 0)$ $P(X = -3) = P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2) = P(X = 3)$. Obtain the probability distribution and the distribution function of X. Answer: <table><tr><td>X</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>$\frac{1}{9}$</td><td>$\frac{1}{9}$</td><td>$\frac{1}{9}$</td><td>$\frac{1}{3}$</td><td>$\frac{1}{9}$</td><td>$\frac{1}{9}$</td><td>$\frac{1}{9}$</td></tr><tr><td>F(x)</td><td>$\frac{1}{9}$</td><td>$\frac{2}{9}$</td><td>$\frac{1}{3}$</td><td>$\frac{2}{3}$</td><td>$\frac{7}{9}$</td><td>$\frac{8}{9}$</td><td>1</td></tr></table>	X	-3	-2	-1	0	1	2	3	P(X)	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	F(x)	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{7}{9}$	$\frac{8}{9}$	1
X	-3	-2	-1	0	1	2	3																			
P(X)	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$																			
F(x)	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{7}{9}$	$\frac{8}{9}$	1																			
C	2	The cumulative distribution function (C.D.F.) of a discrete random variable X is given below. $F(x) = \begin{cases} 0 & ; \quad x < 1 \\ \frac{1}{3} & ; \quad 1 \leq x < 4 \\ \frac{1}{2} & ; \quad 4 \leq x < 6 \\ \frac{5}{6} & ; \quad 6 \leq x < 10 \\ 1 & ; \quad x \geq 10 \end{cases}$ Find $P(2 < X \leq 6)$, $P(X = 5)$, $P(X = 4)$, $P(X \leq 6)$, $P(X = 6)$. Answer: $P(2 < X \leq 6) = \frac{1}{2}$, $P(X = 5) = 0$, $P(X = 4) = \frac{1}{6}$, $P(X \leq 6) = \frac{5}{6}$, $P(X = 6) = \frac{1}{3}$																								

C	3	The life in hours of a certain kind of radio tube has the probability density $f(x) = \begin{cases} \frac{100}{x^2} & ; \quad x \geq 100 \\ 0 & ; \quad \text{elsewhere} \end{cases}.$ (1) Find the distribution function and use it to determine the probability that the life of tube is more than 150 hrs. (2) What is the probability that a tube will last less than 200 hrs. if it is known that the tube is still functioning after 150 hrs. of service? Answer: (1) $F(x) = \begin{cases} 1 - \frac{100}{x} & ; \quad x \geq 100 \\ 0 & ; \quad \text{elsewhere} \end{cases}, \quad P(x > 150) = \frac{2}{3}$ (2) $P(X < 200 X > 150) = 0.25$																																							
H	4	Two dies are rolled. Let X denotes the random variable which counts the total number of points on the upturned faces, construct a table giving the non – zero values of the probability mass function. Also find the distribution of X. Answer: <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>P(X)</td><td>0</td><td>$\frac{1}{36}$</td><td>$\frac{2}{36}$</td><td>$\frac{3}{36}$</td><td>$\frac{4}{36}$</td><td>$\frac{5}{36}$</td><td>$\frac{6}{36}$</td><td>$\frac{5}{36}$</td><td>$\frac{4}{36}$</td><td>$\frac{3}{36}$</td><td>$\frac{2}{36}$</td><td>$\frac{1}{36}$</td></tr><tr><td>F(x)</td><td>0</td><td>$\frac{1}{36}$</td><td>$\frac{3}{36}$</td><td>$\frac{6}{36}$</td><td>$\frac{10}{36}$</td><td>$\frac{15}{36}$</td><td>$\frac{21}{36}$</td><td>$\frac{26}{36}$</td><td>$\frac{30}{36}$</td><td>$\frac{33}{36}$</td><td>$\frac{35}{36}$</td><td>$\frac{36}{36}$</td></tr></table>	X	1	2	3	4	5	6	7	8	9	10	11	12	P(X)	0	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	F(x)	0	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$
X	1	2	3	4	5	6	7	8	9	10	11	12																													
P(X)	0	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$																													
F(x)	0	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$																													

Unit – 2 Probability

T	5	The cumulative distribution function (C.D.F.) of a discrete random variable X is given below. <table><tr><td>X</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>F(x)</td><td>0.08</td><td>0.2</td><td>0.4</td><td>0.65</td><td>0.8</td><td>0.9</td><td>1</td></tr></table> Find probability distribution of X, $P(-2 \leq X \leq 1)$ and $P(X \geq 1)$. Answer: <table><tr><td>X</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.08</td><td>0.12</td><td>0.2</td><td>0.25</td><td>0.15</td><td>0.1</td><td>0.1</td></tr></table> $P(-2 \leq X \leq 1) = 0.72$, $P(X \geq 1) = 0.35$	X	-3	-2	-1	0	1	2	3	F(x)	0.08	0.2	0.4	0.65	0.8	0.9	1	X	-3	-2	-1	0	1	2	3	P(X)	0.08	0.12	0.2	0.25	0.15	0.1	0.1
X	-3	-2	-1	0	1	2	3																											
F(x)	0.08	0.2	0.4	0.65	0.8	0.9	1																											
X	-3	-2	-1	0	1	2	3																											
P(X)	0.08	0.12	0.2	0.25	0.15	0.1	0.1																											
T	6	Find the value of k and the distribution function F(x) given the probability density function of a random variable X as, $f(x) = \frac{k}{1+x^2}$; $-\infty < x < \infty$. Answer: $k = \frac{1}{\pi}$, $F(x) = \frac{1}{\pi} \left[\tan^{-1} x + \frac{\pi}{2} \right]$																																
T	7	The probability density function of a continuous random variable X is given by $f(x) = \begin{cases} ax & ; 0 \leq x < 1 \\ a & ; 1 \leq x < 2 \\ 3a - ax & ; 2 \leq x < 3 \\ 0 & ; \text{otherwise} \end{cases}.$ Find the value of a and also find C.D.F of X. Answer: $a = \frac{1}{2}$, $F(x) = \begin{cases} \frac{ax^2}{2} & ; 0 \leq x < 1 \\ ax - \frac{a}{2} & ; 1 \leq x < 2 \\ 3ax - \frac{ax^2}{2} - \frac{5a}{2} & ; 2 \leq x < 3 \\ 0 & ; \text{otherwise} \end{cases}.$																																

T**8**

A cloud service provider records the time (in minutes) required to restore its servers after an unexpected system failure. Let the random variable X denote the server recovery time for a randomly selected failure. Based on historical data, the probability density function of X is given by

$$f(x) = \begin{cases} \frac{x}{200} & ; \quad 0 \leq x \leq 20 \\ 0 & ; \quad \text{elsewhere} \end{cases}$$

The company has promised its customers, through a Service Level Agreement (SLA), that a server will be restored within 15 minutes after a failure.

- (1) Verify that $f(x)$ is a valid probability density function.
- (2) Find the cumulative distribution function (C.D.F.) of X .
- (3) Find the probability that the server is restored within 15 minutes.
- (4) Find the probability that the server recovery time is between 10 and 18 minutes.
- (5) Find the probability that the recovery time exceeds 15 minutes.
- (6) Based on your answers, comment on whether the company is likely to meet its Service Level Agreement (SLA). Justify your answer.

Answer: (1) $f(x)$ is PDF.

$$(2) F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{400}, & 0 \leq x \leq 20 \\ 1, & x > 20 \end{cases}$$

(3) 0.5626 (4) 0.56 (5) 0.4375

Unit – 2 Probability

T

9

The probability distribution and cumulative distribution function of a random variable X are partially given below.

Complete the missing entries in the table.

X	-2	-1	0	1	2	3	4
P(X)	$\frac{1}{20}$	?	$\frac{3}{20}$	?	?	$\frac{1}{5}$	?
F(X)	$\frac{1}{20}$	$\frac{1}{10}$	?	$\frac{2}{5}$	?	$\frac{4}{5}$	1

Answer:

X	-2	-1	0	1	2	3	4
P(X)	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{3}{20}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
F(X)	$\frac{1}{20}$	$\frac{1}{10}$	$\frac{1}{4}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1

***** End of the Unit *****